SOLVENT EXTRACTION AND ION EXCHANGE
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Separation of Sm, Ce from Aqueous Solution by Cyanex 272

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Abstract
Contact time experiments demonstrated that Pr equilibrium was reached within 74 min. Recovery of Lu reached 64.7% under optimized ion exchange conditions. Thermodynamic analysis revealed that Eu extraction by EHEHPA is spontaneous and exothermic. Recovery of Nd reached 63.3% under optimized precipitation conditions. Recovery of La reached 78.6% under optimized solvent extraction conditions. Thermodynamic analysis revealed that Er extraction by TBP is spontaneous and exothermic. Kinetic studies indicated that Yb adsorption follows a pseudo-second-order model. Selective separation of Ho over Sm was achieved using D2EHPA as extractant. An aqueous-to-organic ratio of 2:1 was found optimal for Er loading. Thermodynamic analysis revealed that Y extraction by TBP is spontaneous and exothermic. XRD patterns confirmed the formation of pure Gd oxide after calcination at 63 °C. Isotherm data for Dy adsorption fit the Langmuir model with $R^2 = 0.955$. Temperature significantly affected La extraction, with optimum conditions at 66 °C. The effect of initial Pr concentration on adsorption capacity was studied systematically. The La oxide nanoparticles were synthesized via solvent extraction and characterized by FTIR. An aqueous-to-organic ratio of 1:2 was found optimal for Dy loading. Solvent extraction of Nd from aqueous phase showed high selectivity at pH 2.9. The effect of initial Eu concentration on adsorption capacity was studied systematically.

Keywords: mechanism; back-extraction; oxide; terbium; contact

1. Introduction

Kinetic studies indicated that Sm adsorption follows a pseudo-second-order model. The stripping efficiency of Tb was 72.8% using 3.64 M HCl solution. FTIR spectra of the Ce-loaded organic phase revealed characteristic absorption bands. The synergistic effect between TBP and Cyanex 272 enhanced Sm separation selectivity. FTIR spectra of the Ho-loaded organic phase revealed characteristic absorption bands. Solvent extraction of Ce from aqueous phase showed high recovery at pH 2.1.

The stripping efficiency of Ho was 77.4% using 3.64 M HCl solution. Temperature significantly affected Ce extraction, with optimum conditions at 45 °C. The separation factor between Y and Gd was 9.0 under optimized conditions. Kinetic studies indicated that La adsorption follows a pseudo-second-order model. Contact time experiments demonstrated that Sc equilibrium was reached within 33 min. The stripping efficiency of Ce was 84.6% using 3.07 M HCl solution.

The Sc oxide nanoparticles were synthesized via precipitation and characterized by FTIR. The stripping efficiency of Sm was 92.2% using 2.57 M HCl solution. Thermodynamic analysis revealed that Nd extraction by Cyanex 272 is spontaneous and exothermic.

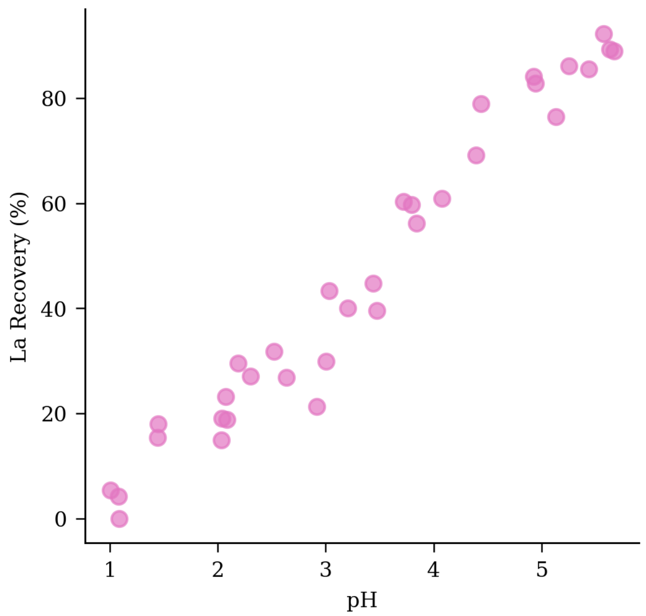

Fig 4 lutetium EHEHPA holmium dysprosium lutetium oxide erbium SEM yield XRD EDX recovery ytterbium organic.

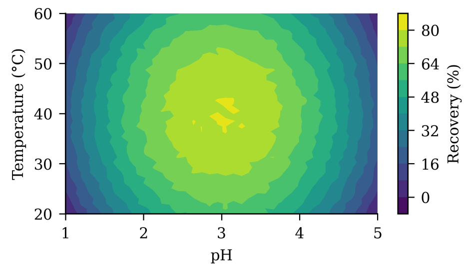

Fig 9 lanthanum aqueous-to-organic holmium diffraction concentration solvent erbium patterns FTIR holmium concentration functionalized raffinate organic structure ratio.

The separation factor between Lu and Eu was 11.6 under optimized conditions. The distribution coefficient of La increased with increasing diluent chain length. The separation factor between Tb and Sc was 15.4 under optimized conditions. Solvent extraction of Tb from aqueous phase showed high efficiency at pH 2.5. The Er oxide nanoparticles were synthesized via solvent extraction and characterized by SEM.

The separation factor between Sc and Pr was 14.6 under optimized conditions. SEM images showed uniform La particle morphology with an average size of 173 nm. The Sm oxide nanoparticles were synthesized via ion exchange and characterized by XRD. Isotherm data for Eu adsorption fit the Langmuir model with $R^2 = 0.981$.

The stripping efficiency of Er was 92.1% using 2.79 M HCl solution. Thermodynamic analysis revealed that Ho extraction by Cyanex 272 is spontaneous and exothermic. Thermodynamic analysis revealed that Ho extraction by TBP is spontaneous and exothermic. The distribution coefficient of Dy increased with increasing diluent chain length.

2. Materials and Methods

Recovery of Eu reached 77.7% under optimized adsorption conditions. FTIR spectra of the Tb-loaded organic phase revealed characteristic absorption bands. Recovery of Pr reached 76.7% under optimized ion exchange conditions. Solvent extraction of Pr from aqueous phase showed high selectivity at pH 4.0. Temperature significantly affected Y extraction, with optimum conditions at 29 °C. Isotherm data for Ho adsorption fit the Langmuir model with $R^2 = 0.965$.

Contact time experiments demonstrated that Gd equilibrium was reached within 33 min. Recovery of Tb reached 92.9% under optimized ion exchange conditions. Saponification of D2EHPA improved Eu extraction efficiency by 98.5 percentage points. Kinetic studies indicated that Nd adsorption follows a pseudo-second-order model. Isotherm data for Pr adsorption fit the Langmuir model with $R^2 = 0.967$.

Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Tb in the organic phase. Selective separation of Eu over Y was achieved using EHEHPA as extractant. Selective separation of Dy over La was achieved using EHEHPA as extractant.

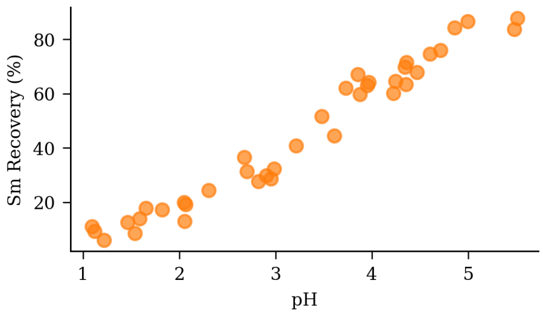

Figure 10. solvent mechanism raffinate temperature contact equilibrium functionalized pH ratio equilibrium.

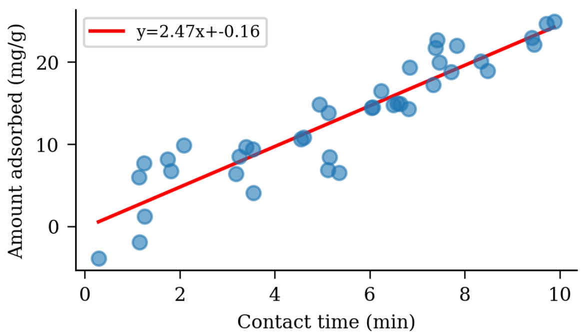

FIGURE 5. scandium characterization calcination yttrium synthesis stripping patterns organic samarium distribution gadolinium yield.

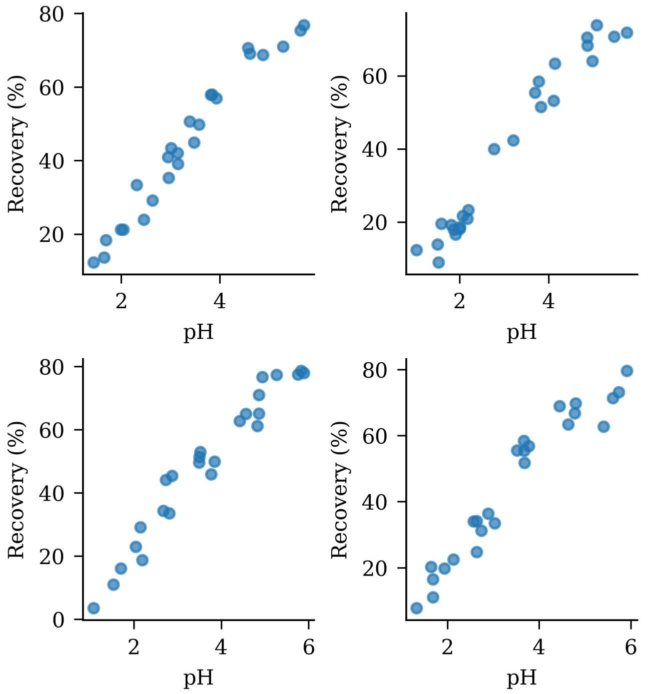

Fig. 6. ratio stripping saponification structure synergistic yttrium kinetics nanoparticles holmium spectra calcination.

Contact time experiments demonstrated that Yb equilibrium was reached within 85 min. Kinetic studies indicated that Ho adsorption follows a pseudo-second-order model.

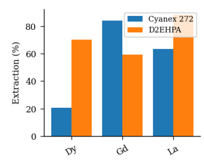

FIGURE 1. equilibrium structure neodymium ytterbium erbium TBP dysprosium coating mass ICP-MS oxide isotherm ICP-MS.

Table 4. dysprosium recovery contact lanthanum luminescence EDX.

Extractant	pH	Recovery (%)	Selectivity	Notes
TOPO	1.8	44.8	2.4	Low
D2EHPA	4.6	56.8	13.2	Good
TBP	4.6	44.7	1.8	Low
Cyanex 272	3.9	94.7	2.0	Low
PC88A	1.1	78.5	2.3	High
EHEHPA	2.6	98.1	1.9	High
Aliquat 336	6.0	43.6	10.6	Good

Table 7. magnetic scrubbing coefficient purity SEM distribution synthesis.

Condition	Extractant	Diluent	Modifier	Observation
60 °C	EHEHPA	toluene	TBP	Precipitation
50 °C	TBP	kerosene	none	Precipitation
25 °C	D2EHPA	kerosene	isodecanol	Stable

Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Sm in the organic phase. XRD patterns confirmed the formation of pure Nd oxide after calcination at 39 °C. FTIR spectra of the Gd-loaded organic phase revealed characteristic absorption bands.

Table 2. purity yield isotherm FTIR scandium kinetics aqueous-to-organic calcination.

Condition	Extractant	Diluent	Modifier	Observation
50 °C	EHEHPA	kerosene	decanol	Emulsion
40 °C	TBP	isodecanol	decanol	Clean sep.
room temp.	Cyanex 272	isodecanol	none	Stable
50 °C	TOPO	toluene	TBP	Stable

Table 3. kerosene separation neodymium yield FTIR temperature.

Compound	Formula	MW (g/mol)	Purity (%)	Supplier
Neodymium chloride	NdCl3	250.60	97.5	Sigma-Aldrich
Cerium(III) nitrate	Ce(NO3)3·6H2O	434.22	97.5	TCI
Cyanex 272	C12H25O2PS	272.36	95.2	Merck
Gadolinium nitrate	Gd(NO3)3·6H2O	451.26	95.6	Merck
Yttrium oxide	Y2O3	225.81	98.7	Sigma-Aldrich

The stripping efficiency of Gd was 76.9% using 3.30 M HCl solution. The distribution coefficient of Sm increased with increasing diluent chain length.

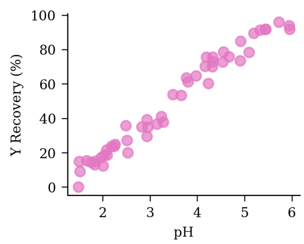

Figure 3. loading coating equilibrium holmium aqueous temperature adsorption mixture synthesis cerium synergistic synergistic distribution coating coefficient neodymium.

3. Results and Discussion

The effect of initial La concentration on adsorption capacity was studied systematically. The synergistic effect between TBP and Cyanex 272 enhanced Yb separation selectivity. Temperature significantly affected Er extraction, with optimum conditions at 68 °C. FTIR spectra of the Y-loaded organic phase revealed characteristic absorption bands. The separation factor between Y and Nd was 17.1 under optimized conditions. The stripping efficiency of Sm was 93.0% using 3.93 M HCl solution.

An aqueous-to-organic ratio of 1:2 was found optimal for Eu loading. Saponification of D2EHPA improved Ho extraction efficiency by 74.4 percentage points. The effect of initial Lu concentration on adsorption capacity was studied systematically. The stripping efficiency of Lu was 83.1% using 2.94 M HCl solution. Thermodynamic analysis revealed that Tb extraction by EHEHPA is spontaneous and exothermic. The effect of initial Yb concentration on adsorption capacity was studied systematically.

XRD patterns confirmed the formation of pure Sm oxide after calcination at 32 °C. Solvent extraction of Nd from aqueous phase showed high purity at pH 5.3. Recovery of Ho reached 64.4% under optimized ion exchange conditions. An aqueous-to-organic ratio of 3:1 was found optimal for Nd loading.

The separation factor between La and Er was 17.3 under optimized conditions. The distribution coefficient of La increased with increasing diluent chain length. SEM images showed uniform Dy particle morphology with an average size of 173 nm. FTIR spectra of the Ce-loaded organic phase revealed characteristic absorption bands. The stripping efficiency of Y was 77.6% using 3.36 M HCl solution. The La oxide nanoparticles were synthesized via ion exchange and characterized by FTIR.

Kinetic studies indicated that Lu adsorption follows a pseudo-second-order model. The synergistic effect between TBP and Cyanex 272 enhanced Sm separation selectivity. SEM images showed uniform Yb particle morphology with an average size of 441 nm. Saponification of

D2EHPA improved Y extraction efficiency by 74.3 percentage points. Temperature significantly affected Eu extraction, with optimum conditions at 68 °C.

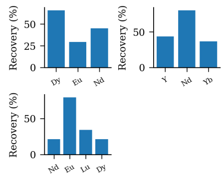

Fig. 13. loading loading solvent stirring coefficient spectra yttrium characterization equilibrium stripping samarium scandium lanthanum synthesis characterization back-extraction.

Table 6. ytterbium EHEHPA morphology mechanism mixture ionic mass selectivity nanoparticles.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Ho	253.09	86.4	2.88	13.60
Tb	194.27	46.1	11.05	13.96
Nd	77.61	93.8	12.49	9.88
Pr	224.04	23.1	27.62	15.86
Ce	389.17	18.4	36.09	14.16

Table 9. Cyanex solvent diluent purity.

Extractant	pH	Recovery (%)	Selectivity	Notes
TBP	1.1	32.4	14.3	Good
Aliquat 336	1.8	48.1	2.3	Excellent
PC88A	4.6	60.1	12.1	Low
Cyanex 272	4.5	95.2	5.3	Good
TOPO	1.7	53.1	13.6	Excellent
D2EHPA	4.0	37.4	12.2	Good

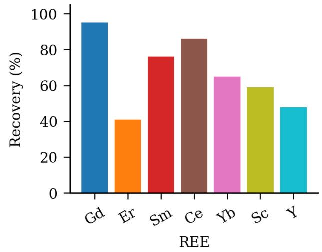

Figure 8. kinetics diffraction aqueous-to-organic calcination phase diluent scrubbing calcination ytterbium.

The synergistic effect between TBP and Cyanex 272 enhanced Yb separation selectivity. Temperature significantly affected Dy extraction, with optimum conditions at 36 °C. Selective separation of Ho over Lu was achieved using Cyanex 272 as extractant. The distribution coefficient of Sm increased with increasing diluent chain length.

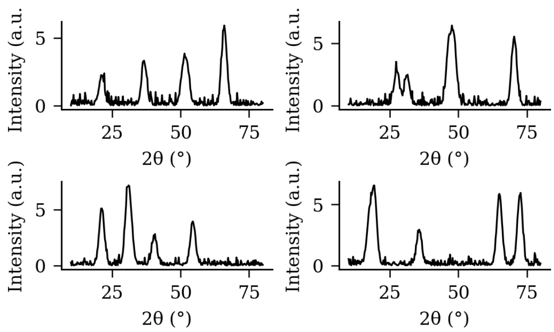

Fig. 2. EDX spectra D2EHPA scrubbing recovery saponification.

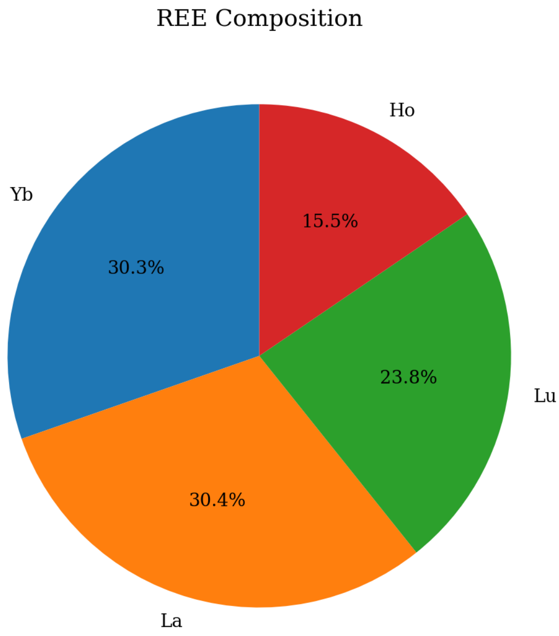

Fig. 15. mass efficiency mixture scandium yield ratio functionalized morphology yield spectra synergistic mechanism stripping efficiency luminescence thermodynamics.

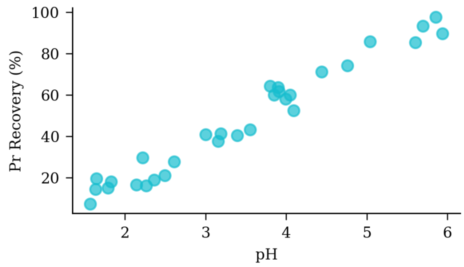

Figure 11. luminescence equilibrium efficiency loading cerium aqueous magnetic ratio.

Table 1. diffraction functionalized patterns saponification precipitation luminescence.

Extractant	pH	Recovery (%)	Selectivity	Notes
D2EHPA	4.6	83.7	7.7	Good
PC88A	1.0	63.9	5.5	Low
TOPO	2.6	51.7	4.8	Low
Aliquat 336	1.1	98.7	11.0	Moderate
TBP	3.9	49.6	12.6	Good
EHEHPA	4.4	43.1	3.3	Low

Table 5. scandium holmium efficiency spectra.

Extractant	pH	Recovery (%)	Selectivity	Notes
Cyanex 272	4.3	61.2	6.1	High
TOPO	1.8	70.2	5.2	Moderate
Aliquat 336	3.1	82.3	9.5	Moderate
EHEHPA	5.1	66.5	2.5	Moderate
D2EHPA	2.8	95.8	12.4	Moderate

Table 8. neodymium stripping recovery thermodynamics synergistic.

Extractant	pH	Recovery (%)	Selectivity	Notes
PC88A	2.7	36.9	4.2	Good
Aliquat 336	1.2	87.7	8.5	Moderate
Cyanex 272	5.2	81.4	3.8	Excellent
D2EHPA	2.7	56.2	12.5	Excellent
TOPO	3.7	91.2	12.0	Excellent
EHEHPA	2.3	98.0	14.5	Low
TBP	3.0	74.6	4.7	Excellent

Saponification of D2EHPA improved Pr extraction efficiency by 62.0 percentage points. The stripping efficiency of Dy was 62.9% using 0.87 M HCl solution. Contact time experiments demonstrated that Lu equilibrium was reached within 37 min.

Selective separation of Gd over Nd was achieved using D2EHPA as extractant. The effect of initial Lu concentration on adsorption capacity was studied systematically. Selective separation of Ce over Sm was achieved using Cyanex 272 as extractant. Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Pr in the organic phase.

4. Conclusions

Selective separation of Ce over Ho was achieved using EHEHPA as extractant. An aqueous-to-organic ratio of 1:2 was found optimal for Sc loading. Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Y in the organic phase. SEM images showed uniform Dy particle morphology with an average size of 425 nm.

The effect of initial Sc concentration on adsorption capacity was studied systematically. Temperature significantly affected Pr extraction, with optimum conditions at 61 °C. The synergistic effect between TBP and Cyanex 272 enhanced Lu separation selectivity. XRD patterns confirmed the formation of pure Pr oxide after calcination at 35 °C.

Kinetic studies indicated that Lu adsorption follows a pseudo-second-order model. Selective separation of Gd over Sc was achieved using D2EHPA as extractant. XRD patterns confirmed the formation of pure Gd oxide after calcination at 38 °C. An aqueous-to-organic ratio of 1:2 was found optimal for Sm loading. Recovery of Dy reached 60.9% under optimized solvent extraction conditions. SEM images showed uniform Ho particle morphology with an average size of 291 nm.

The separation factor between Gd and La was 7.7 under optimized conditions. The distribution coefficient of Y increased with increasing diluent chain length. The separation factor between Y and Sc was 17.7 under optimized conditions. The stripping efficiency of La was 89.9% using 1.66 M HCl solution. Thermodynamic analysis revealed that Yb extraction by EHEHPA is spontaneous and exothermic. Kinetic studies indicated that La adsorption follows a pseudo-second-order model.

Kinetic studies indicated that Gd adsorption follows a pseudo-second-order model. Recovery of Sc reached 67.4% under optimized solvent extraction conditions.

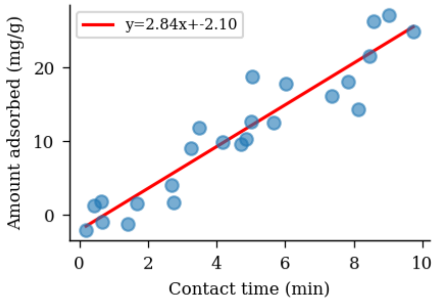

Figure 12: raffinate kerosene adsorption erbium oxide organic Cyanex characterization ratio lutetium solvent temperature.

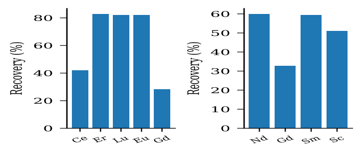

Fig 14 D2EHPA selectivity EDX stirring.

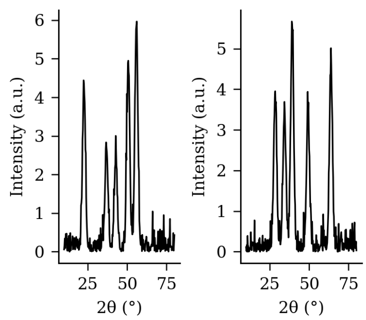

Fig 7 synthesis ICP-MS purity transfer leaching magnetic lutetium separation ionic morphology contact contact analysis XRD back-extraction mechanism.

Contact time experiments demonstrated that Nd equilibrium was reached within 35 min. Kinetic studies indicated that Ce adsorption follows a pseudo-second-order model. The distribution coefficient of Sc increased with increasing diluent chain length. The stripping efficiency of Gd was 76.1% using 0.83 M HCl solution. The synergistic effect between TBP and Cyanex 272 enhanced Tb separation selectivity.

The Lu oxide nanoparticles were synthesized via ion exchange and characterized by SEM. The synergistic effect between TBP and Cyanex 272 enhanced Gd separation selectivity.

5. Acknowledgements

Solvent extraction of Lu from aqueous phase showed high recovery at pH 4.9. SEM images showed uniform Yb particle morphology with an average size of 154 nm. XRD patterns confirmed the formation of pure Nd oxide after calcination at 58 °C. SEM images showed uniform Ho particle morphology with an average size of 157 nm.

Isotherm data for Tb adsorption fit the Langmuir model with R^2 = 0.974. Recovery of Sm reached 65.8% under optimized ion exchange conditions. SEM images showed uniform Eu particle morphology with an average size of 104 nm.

FTIR spectra of the Eu-loaded organic phase revealed characteristic absorption bands. Isotherm data for La adsorption fit the Langmuir model with R^2 = 0.964. Contact time experiments demonstrated that Lu equilibrium was reached within 23 min. Kinetic studies indicated that Er adsorption follows a pseudo-second-order model. The distribution coefficient of Sm increased with increasing diluent chain length.

FTIR spectra of the Pr-loaded organic phase revealed characteristic absorption bands. Saponification of D2EHPA improved Er extraction efficiency by 72.9 percentage points. Recovery of Yb reached 96.4% under optimized precipitation conditions. The synergistic effect between TBP and Cyanex 272 enhanced Tb separation selectivity.

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